OPENING | SLIDE 1

Modern structural bioinformatics

From sequence to testable hypotheses

One sequence. Several levels of inference.

Fold

What structure might it adopt?

Relatives

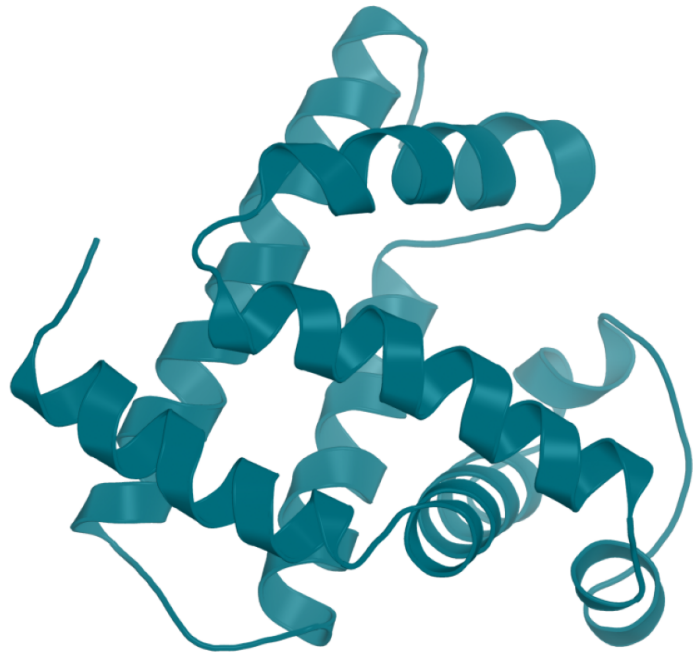
What does structural similarity add?

Interactions

Which molecular arrangements are plausible?

Experiments

What would distinguish competing explanations?



Structural illustration: myoglobin, 1MBN

Keep one protein from your own research in mind.

Coordinate illustration: https://files.rcsb.org/download/1MBN.pdb

Scope: protein structure, structural comparison and molecular interactions; later perspectives on ensembles and design.

REASONING FRAMEWORK | SLIDE 2

A structure is evidence, not an answer.

Keep observations, predictions and hypotheses distinct.

Experimental observation | **Computational prediction** | **Biological hypothesis**

01
Question
Define the claim you want to test.

→

02
Evidence
Retrieve existing knowledge.

→

03
Retrieve or predict
Choose inputs for the question.

→

04
Assess
Inspect uncertainty and plausibility.

→

05
Compare
Seek independent support or conflict.

→

06
Test
Design a new experiment.

New evidence changes the question.

Model output

A cavity is present in the model.

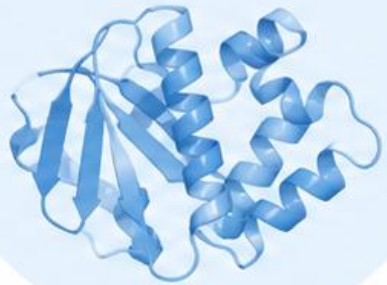
Biological hypothesis

That cavity determines specificity.

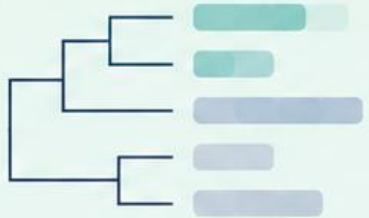
Instructional schematic. The two statements illustrate different levels of inference, not results for a named protein.

What can we infer from a protein sequence?

What fold?

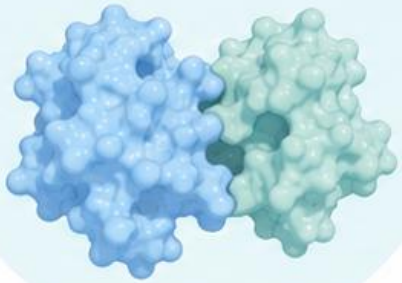


Related to what?

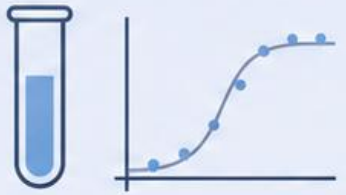


M A T K V L I S L L A N G D V T I Q E L G K
A F D V R D L A A K K A G A E V T L S K G
V T D L V N G K T I V D S L K E A L K A A
• • •

Interacts with what?



What should we test?



Keep one protein from your own research in mind.

Evolution leaves structural clues in sequences.

Conservation and covariation ask different questions.

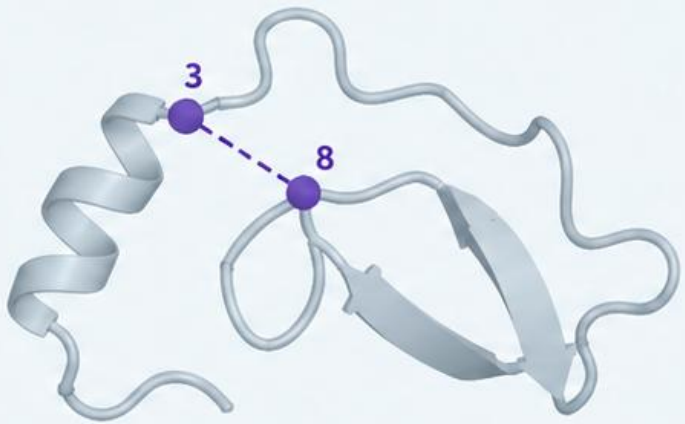
Illustrative alignment

Hand-built teaching example; not a database alignment.

	1	2	3	4	5	6	7	8	9	10	11	12
Seq A	M	A	D	G	V	T	I	K	E	L	G	A
Seq B	M	A	D	G	V	T	I	K	E	L	G	A
Seq C	M	A	E	G	V	T	I	R	E	L	G	A
Seq D	M	A	E	G	V	T	I	R	E	L	G	A
Seq E	M	A	D	G	V	T	I	K	E	L	G	A
Seq F	M	A	E	G	V	T	I	R	E	L	G	A

Infer

Candidate constraint



Hypothetical contact, not a computed structure.

Conservation

What tends to stay the same?

Covariation

Which changes occur together?

EVcouplings

Turn sequence information into testable structural hypotheses.

Method context: https://evcouplings.org/

All figures, names, and sequences on this slide are didactic schematics. No EVcouplings analysis is claimed.

EVOLUTIONARY COUPLINGS | WORKED DHFR EXAMPLE

Evolution leaves a record of 3D contacts.

E. coli DHFR | 159 residues | 3,616 aligned sequences | effective depth ≈ 1,568

1 Read a protein family

	13		121
DYR_ECOLI	V D R **V** I G M	⋯	E V E **G** D T H
K2QU68	L N N **G** I G I	⋯	S F D **C** D A F
Q2B1Y2	E N S **A** I G K	⋯	K F E **A** D T F
UPI0002897BAD	A N R **L** I G K	⋯	Q F D **G** D R Y
UPI00026C4D6F	D N R **I** I G K	⋯	E I Q **G** D T Q
A6EGV0	E D H **G** I G R	⋯	S Y E **A** D T F
K2GJ55	D N G **C** I G R	⋯	A V D **G** D A F
F7IFS4	Q N M **G** I G K	⋯	D F E **S** D T F

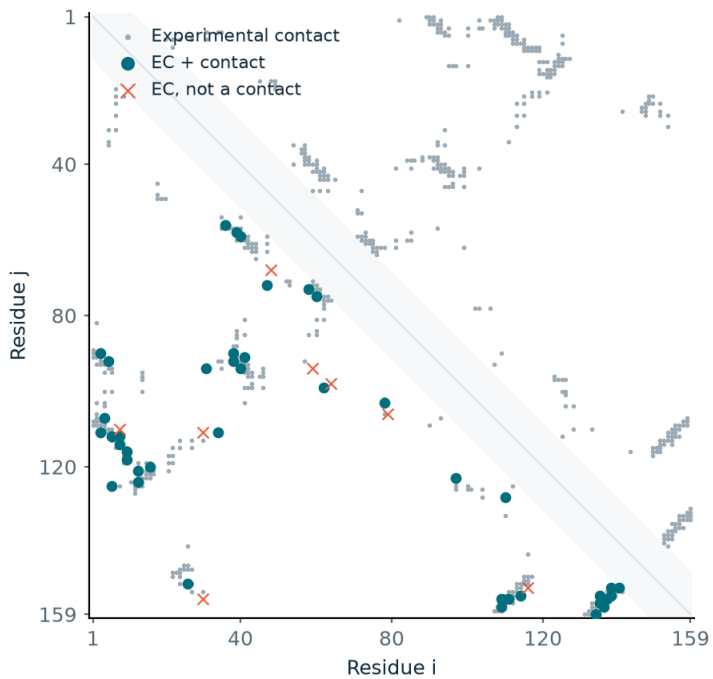
Eight illustrative rows; all 3,616 used for inference.

Changes are not always independent.



108 positions apart in sequence.

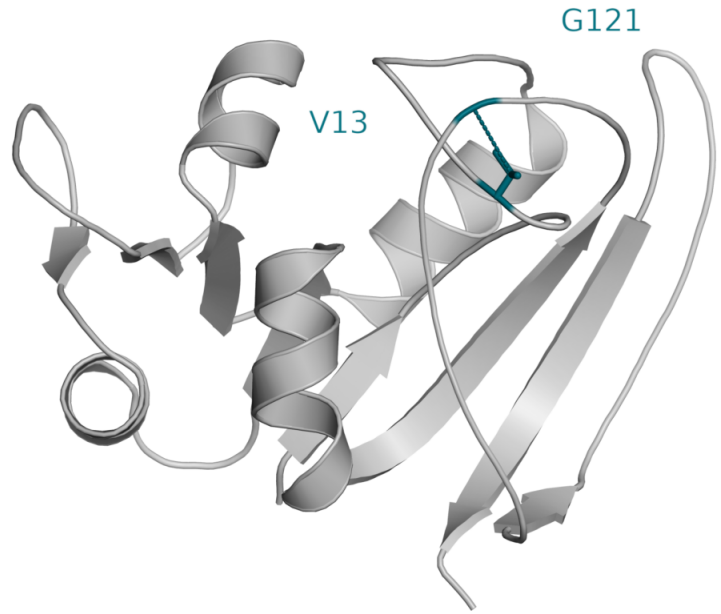
2 Infer residue-pair constraints



42 / 50

top long-range pairs meet the contact rule.

3 Check an experimental fold



V13 / G121 5.05 Å

Far in sequence. Close in space.

The structure was not an inference input.

A coupling score is not a contact probability, a causal mutation effect, or a guarantee of physical contact.

Demo: 100 L-BFGS iterations (iteration cap reached). Contact: Cβ, or Gly Cα, <8 Å; |i−j| ≥12. 1RX2 A has D37 versus focus N37.

Alignment / inference: https://github.com/debbiemarkslab/plmc

Experimental reference: https://www.rcsb.org/structure/1RX2

Learning the language of proteins.

A learned representation is not yet a structural answer.

Many protein sequences

MALW...KQAV

MVLG...DLEK

MGSA...RVAA

MKKI...LNAD

Learn

Sequence representation

ESMC



Conceptual feature display; not actual embeddings.

Optional MSA input

Structure prediction

ESMFold2

Representation → 3D model

Other downstream analyses

Depend on the task and model.

Ask: what was learned, what is being predicted, and how was it validated?

Protein language model ≠ structure predictor

Model distinctions: https://github.com/Biohub/esm

ESMC supplies learned representations; ESMFold2 combines representations with a structure-prediction architecture and supports optional MSA input.

One sequence. A residue distribution. Variant scores.

E. coli DHFR | 159 residues | ESM-2, 650M parameters | frozen pretrained model

1 Hide one residue

Sequence context around position 27

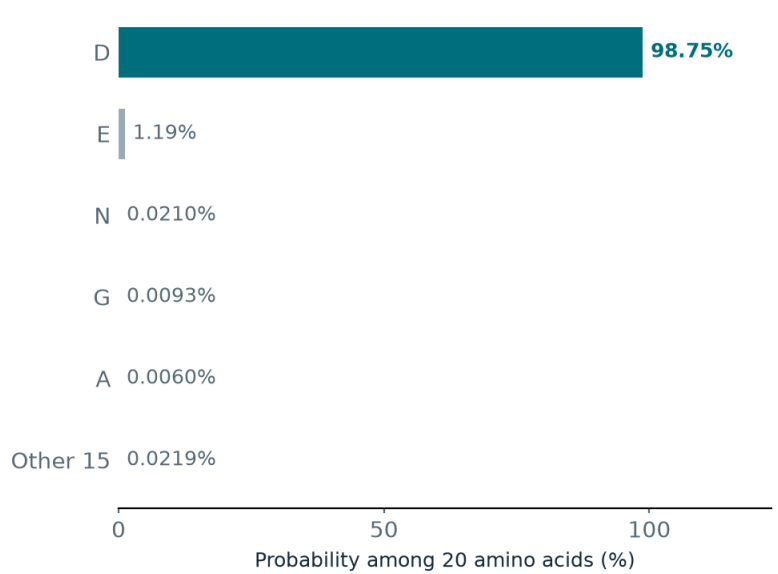
N	L	P	A	?	L	A	W	F
23	24	25	26	27	28	29	30	31

The target token is replaced with <mask>.

One native sequence.
One residue hidden.
No alignment submitted.

No DHFR-specific training in this demo.

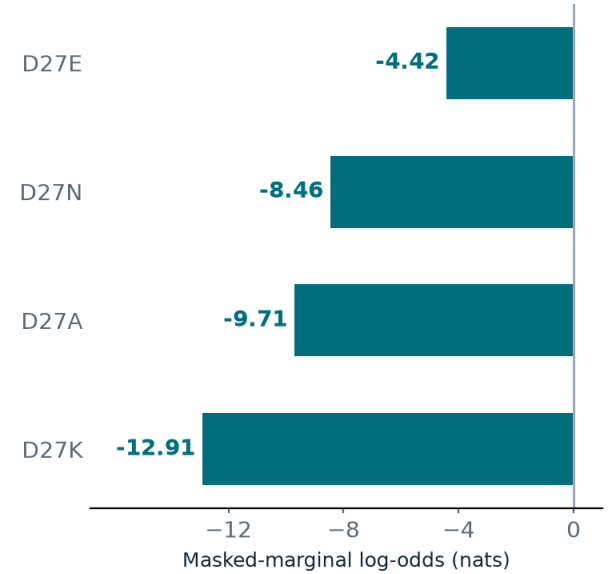
2 Read the model distribution



Asp (D): 98.75%

A conditional model distribution,
not an activity probability.

3 Compare changes with native



D27E: ln(P(E) / P(D)) = -4.42

0 = native residue baseline.
More negative = less sequence support.

Sequence plausibility can help prioritize experiments. It is not a measurement of activity, stability or affinity.

Single masked site in the otherwise native sequence; no fine-tuning, MSA or coordinates. Display probabilities renormalized over 20 standard amino acids.

Model: https://huggingface.co/facebook/esm2_t33_650M_UR50D

Scoring: https://github.com/facebookresearch/esm

Retrieve a model, or make a prediction?

Start from the biological question, then choose the inputs and provenance you need.

Retrieve existing evidence

Accession → model → provenance → assessment



Human calmodulin-1

P0DP23

AF-P0DP23-F1

Model version 6

Existing prediction, not an experimental structure.

Specify a new prediction

Define the molecular system first.

- Protein only
- Several protein chains
- Protein + cofactor or ligand

AlphaFold 3 / Boltz-2

Coordinates + confidence + recorded inputs

The question determines the modelling task. An interface may still restrict the inputs.

Existing model: https://alphafold.ebi.ac.uk/files/AF-P0DP23-F1-model_v6.cif

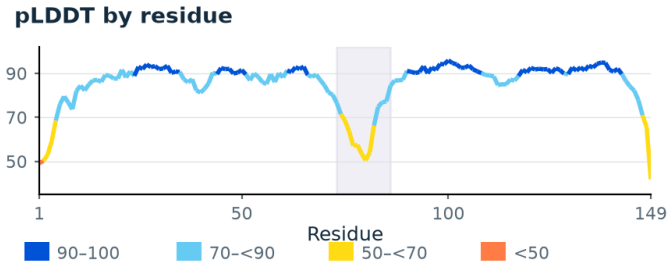
Inputs: https://alphafoldserver.com/guides | https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md

Confident local folds. Uncertain relative placement.

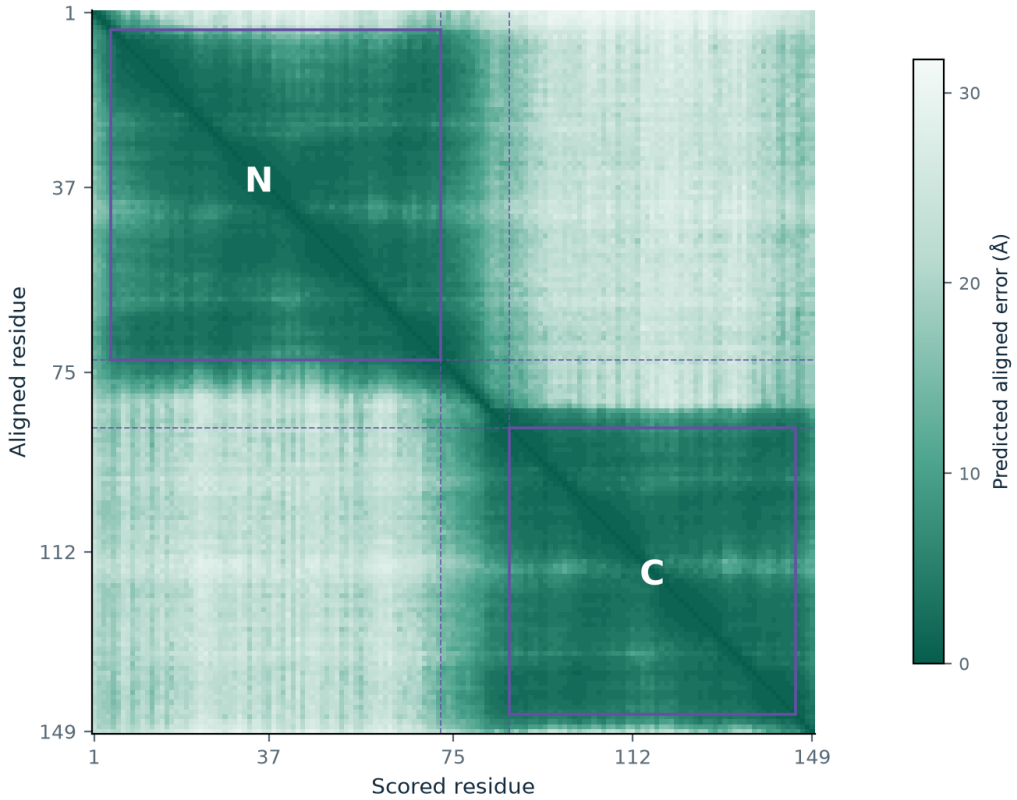
Human calmodulin-1 | P0DP23 | AlphaFold Database model version 6 | 149 residues

A Local confidence





B Relative-position uncertainty



Summary windows: N 5–72; C 87–145. These are analysis windows, not exact domain annotations. PAE is predicted error, not measured motion.

Read the two together

Within-lobe windows

N: 4.0 Å **C: 3.4 Å**

Between windows

22.4 Å mean PAE

A useful local model does not fix the relationship between its lobes.

Model: https://alphafold.ebi.ac.uk/files/AF-P0DP23-F1-model_v6.cif

PAE: https://alphafold.ebi.ac.uk/files/AF-P0DP23-F1-predicted_aligned_error_v6.json | Means calculated from the saved matrix.

A prediction is not a unique biological state.

Sequence-aware comparison, aligned on the N-lobe only



The AFDB model is not assigned to an experimentally established apo state.
This comparison does not measure motion or state populations.

What is being compared?

AlphaFold model

AF-P0DP23-F1, version 6

Experimental reference
1CLL, calcium-bound
Green spheres: calcium ions

Fit uses matched residues in
AF positions 5–72 only.

68 Cα pairs | fit RMSD 1.26 Å

Coordinates: https://alphafold.ebi.ac.uk/files/AF-P0DP23-F1-model_v6.cif | https://files.rcsb.org/download/1CLL.pdb

Sequence mapping and the N-lobe-only transformation are saved with the teaching package. Colours identify provenance, not confidence.

Search broadly. Then choose a comparison worth inspecting.

Actual Foldseek run | 1MBN chain A | 3Di/AA | PDB100, server-reported version 20240101

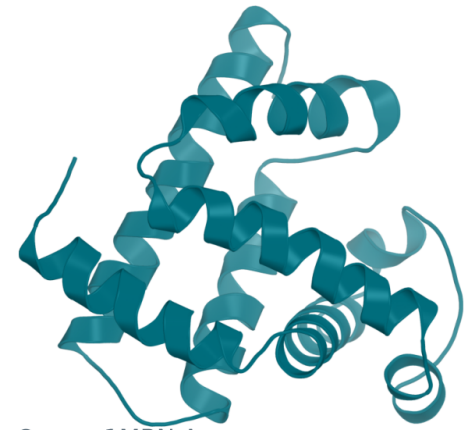
925
returned alignments

153
query residues

899
1FSL B: returned order

Selected rows from the real result list; original positions retained

Order	Target	Seq. ID	Aln. cols	E-value
1	1MBN A | myoglobin (self hit)	100.0%	153	9.4e-22
2	5B84 A | myoglobin I107Y	99.3%	153	3.5e-18
3	1U7S A | myoglobin	100.0%	153	4e-18
218	4B3W B | cytoglobin	30.9%	155	9.5e-09
635	5EET A | neuroglobin	16.8%	148	2.7e-05
761	2LH7 A | leghemoglobin	17.1%	152	0.00023
899	**1FSL B | soybean leghemoglobin**	**12.1%**	**148**	**0.013**



Query: 1MBN A
Protein only; heme omitted

Do not stop at the top hits.
Returned hits are not all confirmed homologues.

A distant structural comparison lies well below the near-identical hits.

Sequence identity and E-values refer to this search alignment; a separate structural comparison follows.

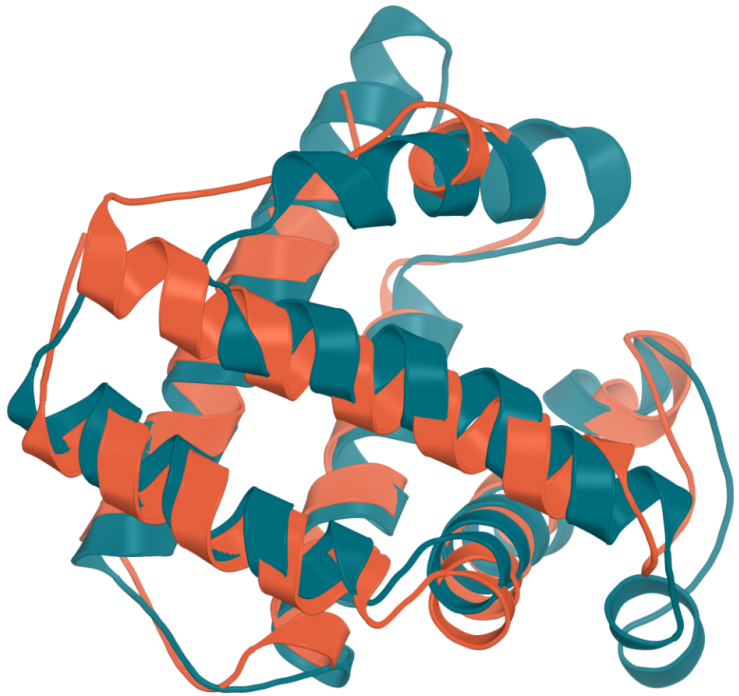
Query coordinates: https://files.rcsb.org/download/1MBN.pdb | Database metadata: https://search.foldseek.com/api/databases

Saved run: 9JUQmuFtzzCi-ufQrRv0mkHE562y671NBHSwsw | Retrieved 21 October 2026 (Europe/Paris); full raw JSON, archive and CSV preserved.

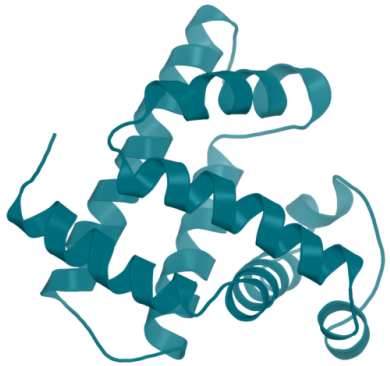
STRUCTURAL COMPARISON | SLIDE 9

A shared structural core despite little sequence identity.

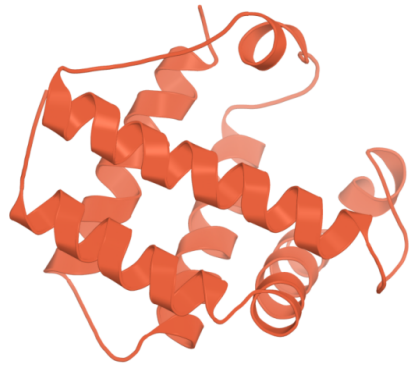
Myoglobin 1MBN A and soybean leghemoglobin 1FSL B | Independent TM-align comparison



Teal: myoglobin | Coral: leghemoglobin
Protein cartoons; heme and other ligands omitted.



1MBN A



1FSL B

0.767
TM-score / query length 153

2.39 Å
Cα RMSD over matched pairs

136
matched residue pairs

11.8%
identity over non-gap pairs

First 76 alignment columns (excerpt)

```
1MBN  V-LSEGEWQLVLHVWAKVEADVAGHGQDILIRLFKSHPETLEKFDRFKHLKTEAEMKASEDLKKHGVTVLTALGAI
      |   |    ||        |                |     |                |
1FSL  VAFTEKQDALVSSSFEAFKANIPQYSVVFYTSILEKAPAAKDLFSFLA-N----GV-DPTN-PKLTGHAEKLFALV
```

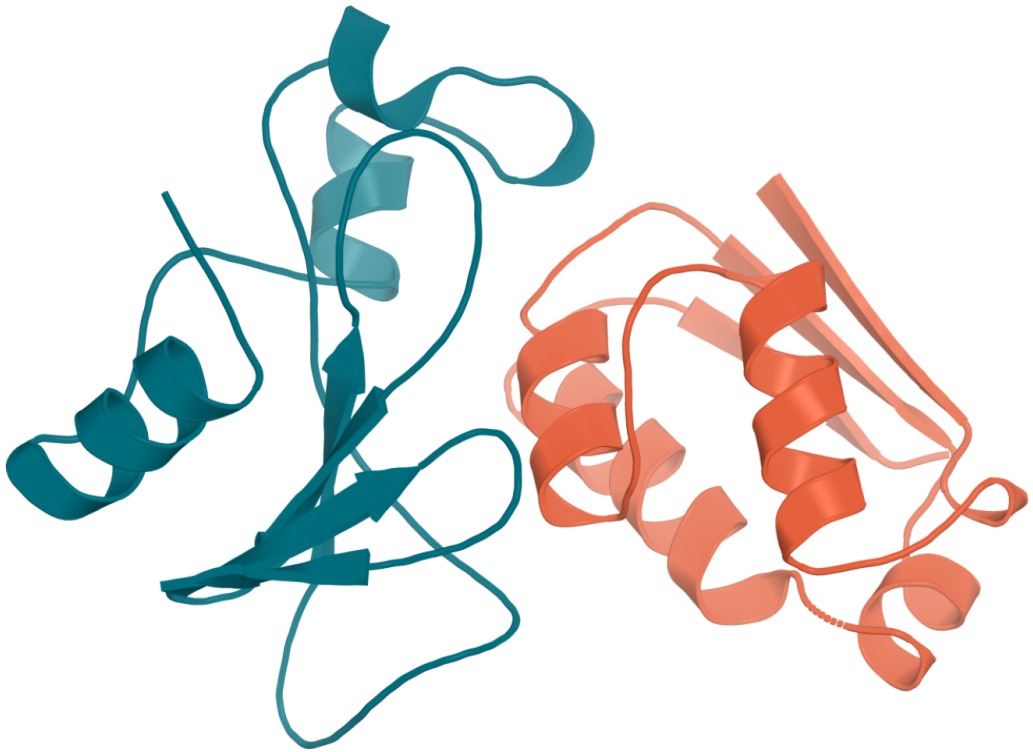
Search identity: 12.1% (server alignment).
TM-align identity above uses a different alignment and an explicit non-gap denominator.

Coordinates: https://files.rcsb.org/download/1MBN.pdb | https://files.rcsb.org/download/1FSL.pdb

TM-align through Foldseek --alignment-type 1; protein-only chain B matches the returned hit. Exact pairs, scores and transformation saved.

From one chain to an assembly.

Define the components and stoichiometry, then assess their arrangement.



1BRS, biological assembly 1 | Experimental reference

Teal: barnase A | Coral: barstar D

Start with specified components

1 × barnase + **1 × barstar**

Composition

Which chains, and how many copies?

Interface

Inter-chain confidence, contacts, clashes.

Biology

Independent evidence for interaction.

A plausible interface is not proof of a physiological interaction.

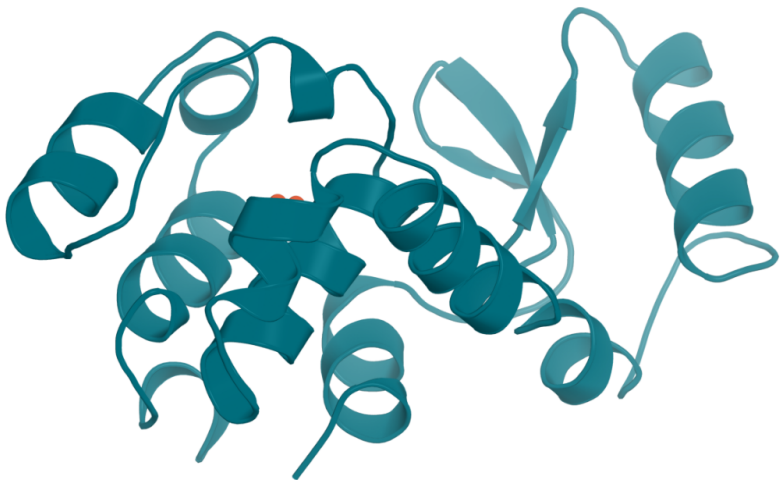
Experimental coordinates: https://files.rcsb.org/download/1BRS.pdb

Interface-confidence context: https://alphafoldserver.com/guides | This view is not a new AlphaFold or Boltz prediction.

From protein and molecule to a binding hypothesis.

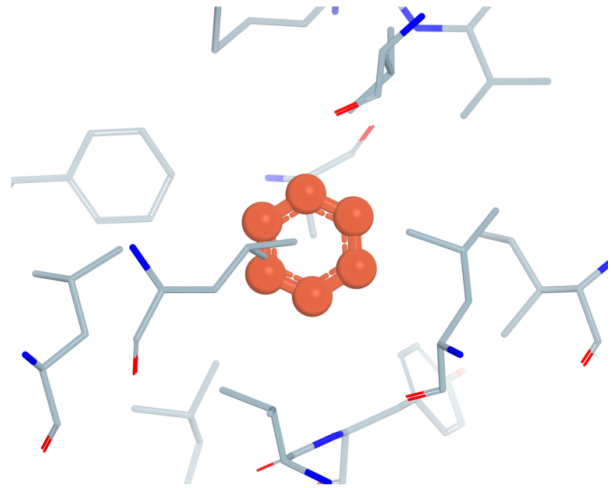
Specify the molecular components, then evaluate the predicted arrangement.

Experimental reference



T4 lysozyme L99A | 4W52

Experimental site view



Benzene: coral | Nearby protein atoms

A modelling workflow

01 Inputs

Deposited protein construct
+ benzene: c1ccccc1

02 Prediction

AlphaFold 3 / Boltz-2
Coordinates + confidence

03 Assessment

Location, contacts, clashes,
alternative poses

Structural confidence ≠ binding affinity ≠ catalytic activity

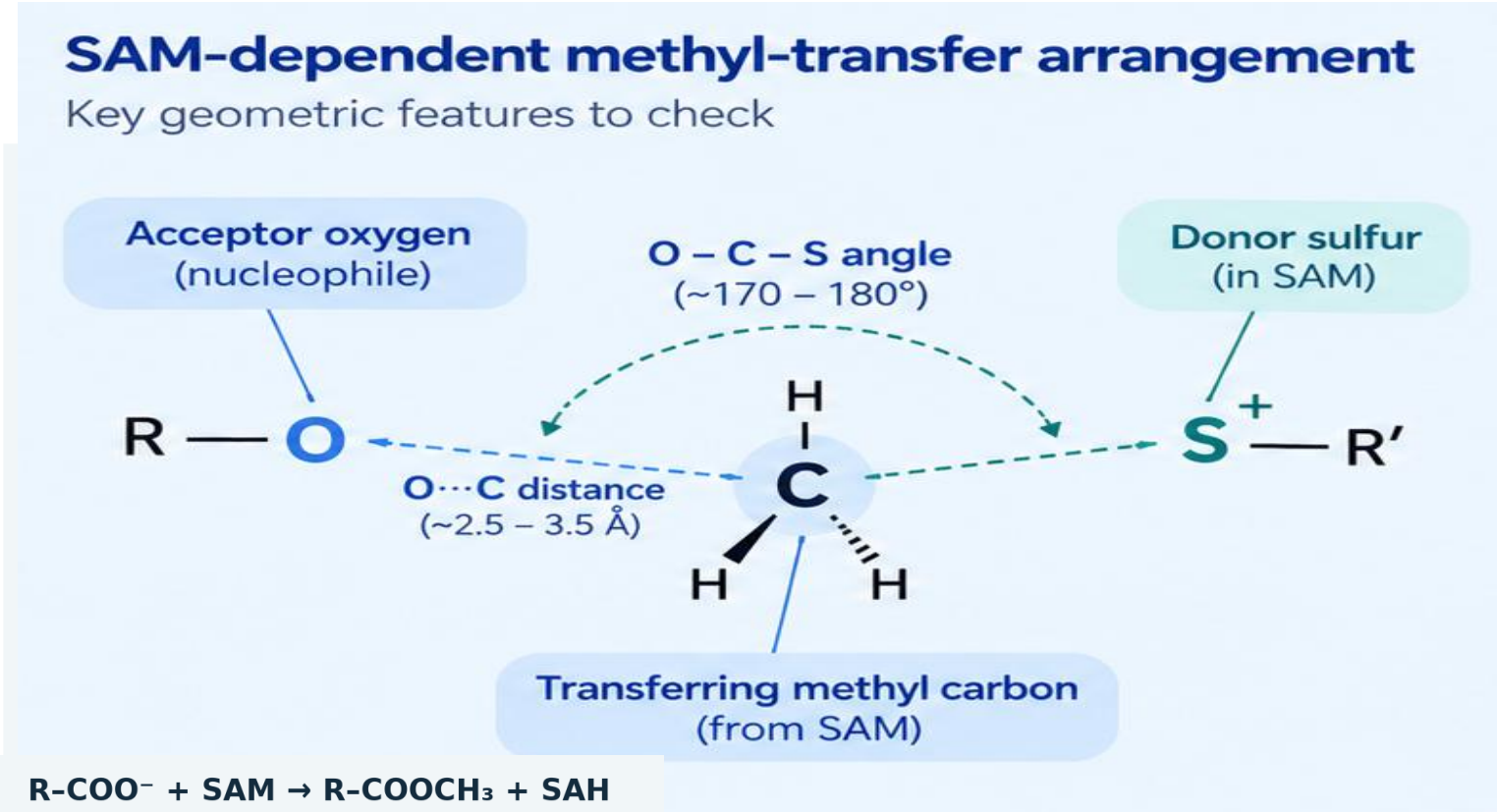
Reference coordinates: https://files.rcsb.org/download/4W52.pdb | Benzene is the crystallographic ligand BNZ.

Input support and confidence/affinity outputs: https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md

The molecular views are experimental references. The workflow is schematic; no new predicted pose or affinity is claimed.

Binding plausibility is not catalytic competence.

Ask whether the reacting atoms can adopt a chemically sensible arrangement.



Geometry as a check

Distance

Acceptor O to methyl C

Direction

O–C–S angle at methyl C

Also inspect

Charge and protonation
Stereochemistry and clashes
Alternative poses

In-line attack is a chemical hypothesis, not a universal numerical cutoff.

A static pose alone does not establish transition-state access, mechanism or reaction rate.

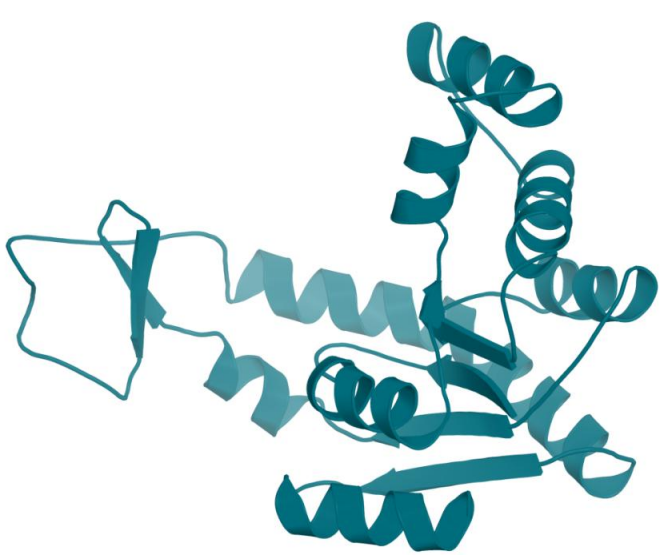
Generic geometry and mechanistic context: https://pmc.ncbi.nlm.nih.gov/articles/PMC6503465/

Idealised explanatory schematic, not an enzyme model or a transition-state calculation. No target-specific geometry or result is shown.

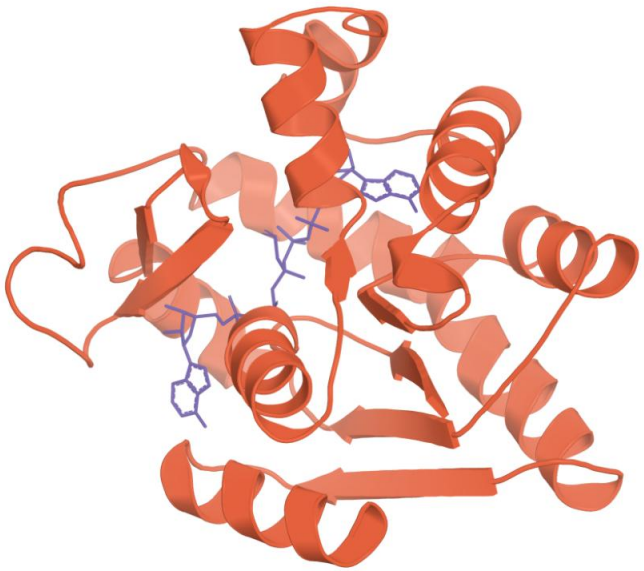
One structure is not an ensemble.

Experimental references illustrate the question; generative sampling addresses a different task.

Adenylate kinase



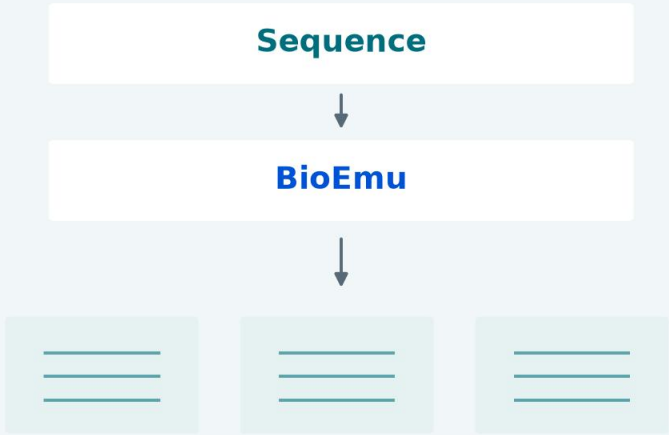
4AKE A | Unligated reference



1AKE A | AP5A-bound reference

Two experimental structures, aligned by matched Cα positions.
They are not BioEmu outputs and are not successive time points.

An equilibrium-sampling question



Independent approximate-equilibrium samples

Output schematic only; no new BioEmu run.

A collection of conformations is not a kinetic trajectory.

Experimental references: https://files.rcsb.org/download/4AKE.pdb | https://files.rcsb.org/download/1AKE.pdb

BioEmu: https://github.com/microsoft/bioemu | https://doi.org/10.1126/science.adv9817

Design reverses the question.

What new protein could satisfy a target, an objective and a set of constraints?



Teal: fixed target | Coral: specified hotspots

Provided target model from the classic RFdiffusion example.
Not an experimental structure or a newly generated binder.

A concrete constraint set

Fixed target

Insulin-receptor fragment A1–150

Specified hotspots

A59, A83, A91

Designable component

New binder chain: 70–100 residues

Generate → **Assess** → **Experiment**

Classic input: https://github.com/RosettaCommons/RFdiffusion/blob/main/examples/design_ppi.sh

RFdiffusion3 adds atom-level chemical-interaction constraints: https://www.ipd.uw.edu/2025/12/rfdiffusion3-now-available/

The target and constraints are real example inputs. No generated candidate, predicted binding score or experimental success is claimed.

SYNTHESIS | SLIDE 15

Build an evidence chain, not a gallery of models.

The question determines the method; this is not a mandatory sequence of software.

Sequence evidence
EVcouplings / ESM

→

Structure
AlphaFold

→

Comparison
Foldseek / alignment

→

Interactions
AF3 / Boltz-2

→

Beyond one model
BioEmu / RFdiffusion

Three questions to carry into every analysis

01 What supports this model?

02 What conclusion does it actually support?

03 What experiment could change the interpretation?

An evidence-based conclusion

2 defensible observations

1 unresolved inference

1 discriminating experiment

Synthesis of the methods discussed. Tool names identify questions and outputs, not guarantees or a performance ranking.
Observations, computational predictions and biological hypotheses should remain distinguishable.

Your investigation begins here.

MnMT4 | UniProt V5X766 | Investigate structural evidence, not a predetermined answer.

Known inputs

Protein

MnMT4, 305 residues | V5X766

Cofactor

S-adenosylmethionine (SAM)

Substrate comparison

3-OPC; optional 3-indoleacrylic acid

Your deliverable

2 defensible observations

1 unresolved inference

1 discriminating experiment

Protein | **Protein + SAM** | **Protein + SAM + substrate**

Which conclusions are defensible, and what evidence is still missing?

Accession and length: https://www.uniprot.org/uniprotkb/V5X766/entry | Biochemical inputs supplied for the practical.

Compare models prepared under different input conditions. No target structure, neighbours, residue correspondences or ligand interpretation is revealed.

References | Prediction and structural comparison

ALPHAFOLD 2 AND CONFIDENCE

Jumper, Evans, Pritzel et al. (2021)

Highly accurate protein structure prediction with AlphaFold

Nature 596, 583–589
https://doi.org/10.1038/s41586-021-03819-2

ALPHAFOLD 3 AND BIOMOLECULAR COMPLEXES

Abramson, Adler, Dunger et al. (2024)

Accurate structure prediction of biomolecular interactions with AlphaFold 3

Nature 630, 493–500
https://doi.org/10.1038/s41586-024-07487-w

BOLTZ-2: STRUCTURE AND AFFINITY

Passaro, Corso, Wohlwend et al. (2025)

Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction

bioRxiv, 2025.06.14.659707 [preprint]
https://doi.org/10.1101/2025.06.14.659707

FOLDSEEK STRUCTURE SEARCH

van Kempen, Kim, Tumescheit et al. (2024)

Fast and accurate protein structure search with Foldseek

Nature Biotechnology 42, 243–246
https://doi.org/10.1038/s41587-023-01773-0

TM-ALIGN STRUCTURAL ALIGNMENT

Zhang and Skolnick (2005)

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